

Creating a Digital Twin and Polar Digital space in Antarctica

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ABSTRACT

Through the creation of a polar digital space and the data architecture in it, an attempt will be made to get as close as possible to the real possibilities of building a Digital Twin in the Antarctic or the polar region, and also to propose the European data policy in the Antarctic and the Arctic.

The data sources and instruments at many of the Antarctic bases are not interoperable, and this creates a major problem for comparison and simulation modeling even on a single island, let alone a larger area or the Antarctic mainland.

In most of the polar summer, the optical data is not of good quality due to the presence of frequent cloud cover. This necessitates the mandatory use of in situ data in order to verify the others.

The purpose of this type of research is to enforce certain standards for data dissemination such as Open Data, Open Science, and also to show the dynamics of certain territories on Livingstone Island based on research from the summer of 2022-2023 through various methods, indices and applications, published on the project page¹. The complex approach used in the research provides reliable, up-to-date and interoperable information, regardless of the geographical location of the objects and is devoid of subjective interference.

The use of data from a drone, ground data from SEKONIC spectrometer, data from NAVAL RSV 421ship², satellites isn't a new method, but a complex. Using the maximum number of applications and also creating an innovative weather station certainly brings even greater added value to society and science, which seriously investigates climate change in polar latitudes.

A ground weather station AWG, powered by an environmentally friendly magnesium-air battery, was developed especially for a project of the Bulgarian Antarctic Base³.

The data and models will serve the Bulgarian initiative for the construction of the Digital Twins in Antarctica and Arctic or Digital polar space, initiatives such as Oceans and Seas, which is being pilot developed in the department of Aerospace Information, Space Research and Technology Institute – Bulgarian Academy of Sciences.

Keywords: Antarctica, Climate changes, Digital polar space, Digital twins, Interoperability, SAR

1. INTRODUCTION

The purpose of this research does not mandate the use or non-use of particular data. This is an attempt to use some of the high-value data that is open access and in machine-readable format for the Antarctic territory.

Combining them with in situ surveys, drone and satellite data, which are shared data on the Internet as a good practice, can create a digital polar space and support initiatives such as Oceans and Seas, Earth's Digital Twins and various types of research with local, regional and global significance.

This research has brought together several use cases that can be used in Antarctic climate change research for different sites of interest.

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Data spaces or data digital spaces are data sharing ecosystems which are built on uniform rules, certified data providers and recipients and trust among partners. They provide a secure and privacy-preserving infrastructure to pool, access, process, use and share data. Data owners have control over who has access to their data, for what purpose and under what conditions it can be used⁴.

The aim of this study is to demonstrate the importance of creating a polar data space that is properly structured and interoperable databases. In this way, in addition to contributing to a more effective study of climate change, they will also contribute to the construction of many different models, one of which may be the Earth's digital twin.

According to the author, the construction of such a space can be upgraded and supplemented, because in addition to distance research, there are other fields that actively generate data in polar conditions, such as medicine, biology, sociology and others.

The fact is, however, that there are also quite a few differences in data structuring, and this is mainly due to the international teams and projects that develop mainly in the Antarctic region, because in the Arctic and in the European part, the data are more clearly defined and correspond of the interoperability requirements.

In this regard, the European Commission rules and Inspire, Open and Spatial Data Policy could easily be proposed and implemented in relation to high-value data⁵ to which the data from research in the Antarctic during the summer season 2022-2023 fall, where a synchronous monitoring approach was made based on remote sensing, Unmanned aerial vehicles(UAV's) and in situ data.

Additionally, as hinted by their name, they are especially valuable when it comes to creating value-added services, applications and, more generally, high-quality jobs for society and the EU⁶, figure 1.

The identification and definition of these high-value datasets lead to a significant change in the field of open data, as the chosen datasets will be made available in harmonised technical standards. This aims to increase their potential for reuse and henceforth their impact⁶.

These characteristics help overcome a series of barriers that often restrict the free circulation of information, such as high-use fees, non-machine-readable content, restrictive licences, poor interoperability or poor accessibility due to scattered data holders⁶.

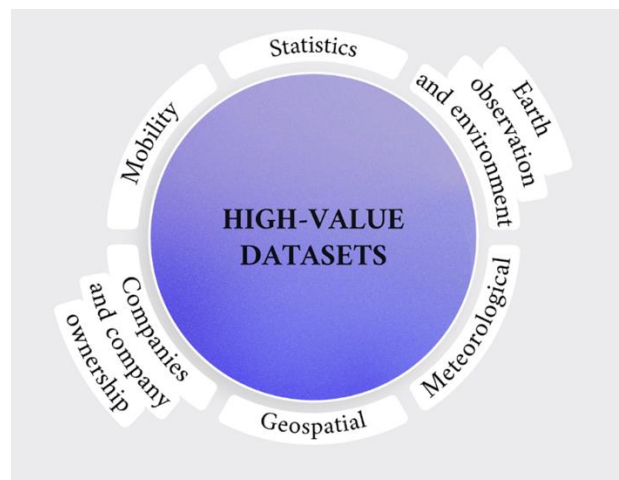


Figure 1 High-value datasets⁶

Thematic Areas, which must necessarily participate in the research and creation of digital space according to the EC, are: Biodiversity, Coastal areas, Disaster risk reduction, Energy, Land use planning, Marine and fisheries, Mountain areas, Tourism, Transport, Water management, Disaster Response, Disaster Recovery and Other.

According to the EC, the Scientific Discipline(s) involved in the construction of the Earth's Digital Twins, and why not in Antarctica, are the following: Solid Earth Geophysics, Biology/ Environmental Science / Ecosystems / Biodiversity, Climatology, Geology, Classical geography, Ecology, Economics, Social sciences, Earth observation, Mathematical modeling, Decision theory and others⁷.

1.1. Region of interest

The area of interest is Livingston Island, near the Bulgarian Antarctic base (BAB), (Fig. 2). The area was selected due to available field research in the southern parts of Livingston Island from 2022. The author has already done remote sensing in the northern part of the island in the period 2016-2018 ^{8, 9}.

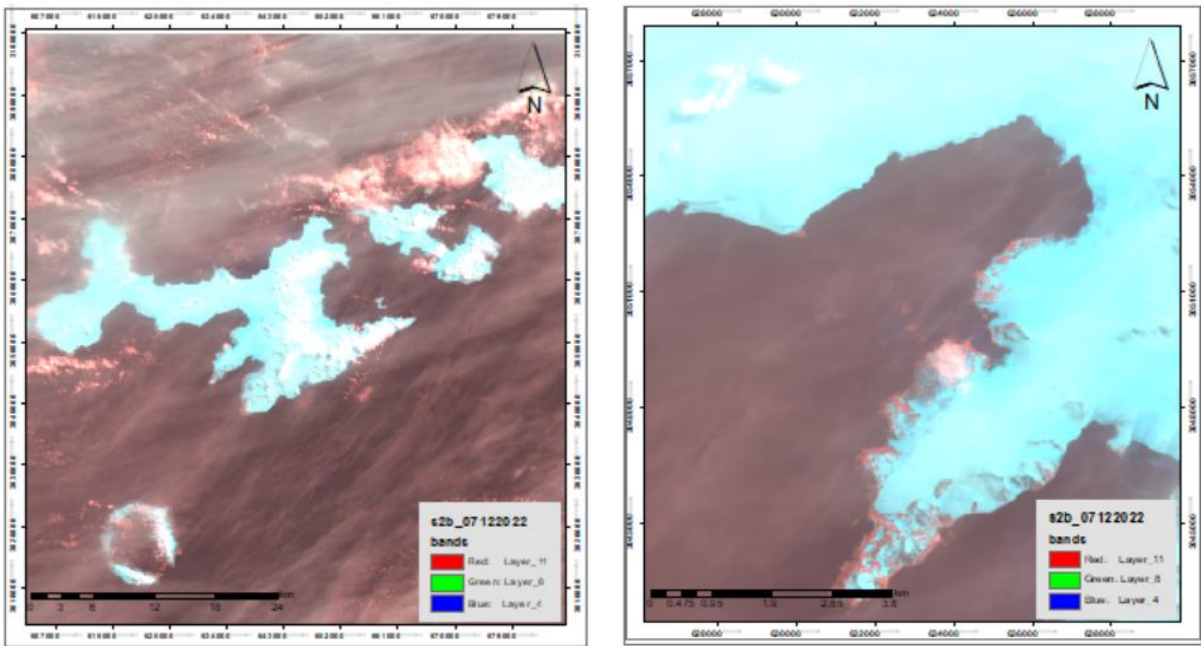


Figure 2 Livingston Island (first) and area around BAB (second) - Sentinel-2MSI composite images (Bands11, 8a, 4)

2. DATA AND METHODS

2.1. Data acquisition

Spectral analysis is one of the most important physical methods for studying objects. It’s used to determine qualitative and quantitative characteristics of objects based on their spectral curve, generated from in situ and satellite data.

Data from Sentinel-2 by the European Space Agency (ESA) as part of the Copernicus program were used. Sentinel-2 is a land-monitoring constellation of two identical polar-orbiting satellites—Sentinel-2A and Sentinel 2B—phased at 180° to each other. It is equipped with a state-of-the-art high-resolution multispectral imager (MSI) instrument, which provides a 290 km swath width and covers all of Earth’s land surfaces, large islands, and inland and coastal waters every five days¹⁰.

Table 1 Image acquisition dates Sentinel 2MSI (ESA)¹¹

SENTINEL 2A MSI	DATE OF ACQUISITION
SENTINEL 2A MSI	07/12/2022
SENTINEL 2A MSI	16/01/2023

When it’s difficult to use cloud-free optical images, it is imperative to apply alternative methods and data for evaluation, such as using satellite data from the microwave range, where the presence of cloud cover is not an obstacle. This stems from the different wavelengths in the optical and microwave ranges, and the use of the microwave range appears to be

more appropriate for the purposes of the study. With the help of radar images, the detection, identification and recognition of water, wet areas, wet snow, ice and rocks is easier and is applicable for polar research^{12, 13}.

Satellite images from the Sentinel-1B and 1A sensors were used. The dates of the satellite images used need to match the rest of the satellites or be as close as possible in order to apply a comprehensive approach and data validation.

From the Sentinel -1 products, the GRD format, type IW, is selected, which usually contains images with hv, vv or hh polarization, but for Antarctica hh polarization is available and used, with a length of $\lambda=5.6\text{cm}$ in C-band from the Electromagnetic spectrum. The spatial resolution of a pixel in the polar region is about 12/16 meters. The 2019 data has a resolution of 33/35 meters (table 2).

Table 2 Image acquisition dates SAR from Livingston island hh polarization, Sentinel 1(ESA)¹¹

Sentinel 1 SAR	DATE OF ACQUISITION
Sentinel 1B SAR hh polarization	04/01/2019
Sentinel 1B SAR hh polarization	03/03/2019
Sentinel 1B SAR hh polarization	30/03/2019
Sentinel 1A SAR hh polarization	08/01/2020
Sentinel 1A SAR hh polarization	01/02/2020
Sentinel 1A SAR hh polarization	08/03/2020
Sentinel 1A SAR hh polarization	01/04/2020
Sentinel 1A SAR hh polarization	11/01/2021
Sentinel 1A SAR hh polarization	07/02/2021
Sentinel 1A SAR hh polarization	03/03/2021
Sentinel 1A SAR hh polarization	08/04/2021
Sentinel 1A SAR hh polarization	09/01/2022
Sentinel 1A SAR hh polarization	02/02/2022
Sentinel 1A SAR hh polarization	10/03/2022
Sentinel 1A SAR hh polarization	03/04/2022
Sentinel 1A SAR hh polarization	16/01/2023
Sentinel 1A SAR hh polarization	09/02/2023
Sentinel 1A SAR hh polarization	05/03/2023
Sentinel 1A SAR hh polarization	10/04/2023

Derivatives are formed during image processing images, which requires the addition of new folders for data and metadata, and it is good that they are saved outside the main ones, which are according to the standard set by the European Space Agency (ESA). The author recommends that resized or georeferenced images be titled the same way, and in this case they are titled Resize for data interoperability and compliance with the project's FAIR data principles (figure 3).

Satellite data from Sentinel 3 have been processed with the SNAP product, which can be freely used and is published on the European Space Agency website¹⁴. All images have been transformed to degrees Celsius for greater reference when analyzing the results.

Satellite data from the Sentinel 3 SLSTR measuring land surface temperature (LST) in the Livingstone Island area provide the information for 5 years (tabela 3).

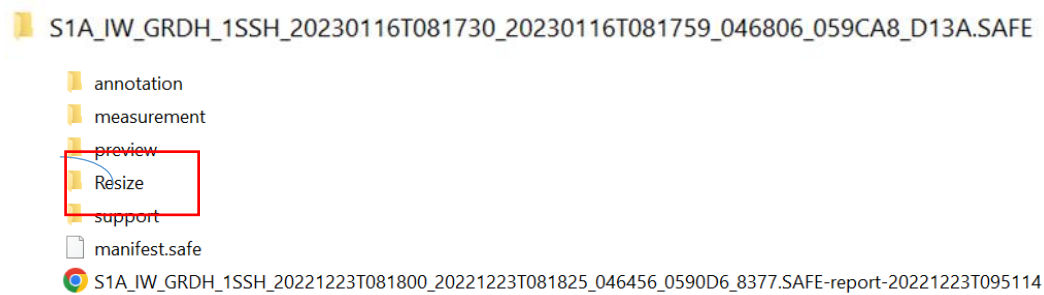


Figure 3 Sentinel 1A image metadata granule¹¹

Table 3 Image acquisition dates Sentinel 3 SLSTR (ESA)^{11, 15}

SENTINEL 3 SLSTR	DATE OF ACQUISITION
Sentinel 3B	07/12/2018
Sentinel 3A	12/12/2018
Sentinel 3A	24/12/2018
Sentinel 3B	07/12/2019
Sentinel 3A	12/12/2019
Sentinel 3A	24/12/2019
Sentinel 3B	07/12/2020
Sentinel 3A	12/12/2020
Sentinel 3A	24/12/2020
Sentinel 3B	07/12/2021
Sentinel 3A	12/12/2021
Sentinel 3A	24/12/2021
Sentinel 3A	07/12/2022
Sentinel 3A	12/12/2022
Sentinel 3B	24/12/2022

The simplistic definition would be how hot the "surface" of the Earth would feel to the touch at a certain location. From the satellite's perspective, the "surface" is all it sees when it looks through the atmosphere at Earth. It can be snow and ice, the grass on the lawn, the roof of a building or the leaves in the crown of the forest. LST is not the same as air temperature, which is included in the daily weather report¹⁶.

The LST retrieval algorithms are accurate to 1 K, especially at night when differential surface heating is absent¹⁶.

Since the study area was tested in the summer when the Sun doesn't set, no significant differences in temperature were observed between morning and midday temperature measurements, as images from two different time ranges were obtained.

Land surface temperature (LST) data are essential for understanding Antarctic climate change and how it will affect urban and rural areas, where human activity and population growth are the main drivers of land surface thermal variation.

Level-2 LST provides land surface parameters with a resolution of 1 km. In addition to a measurement file with Land Surface Temperature (LST) values calculated and provided for each pixel, there are also a number of associated auxiliary parameters that are visible from the product's drop-down menu when it's loaded (figure 4). The figure presents a small part of the parameters that are contained in the granule.

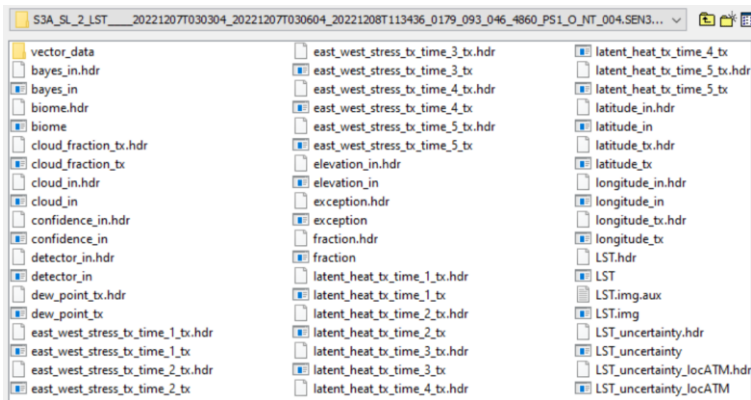


Figure 4 Sentinel 3 SLSTR Level-2 LST^{11, 15}

Altitude and depth of snow cover in the Livingston Island area measured by Level-2 LST were used for data analysis (figure 5 and figure 6)^{11, 15}.



Figure 5 Altitude calculated by elevation in Level-2 LST^{11,15}

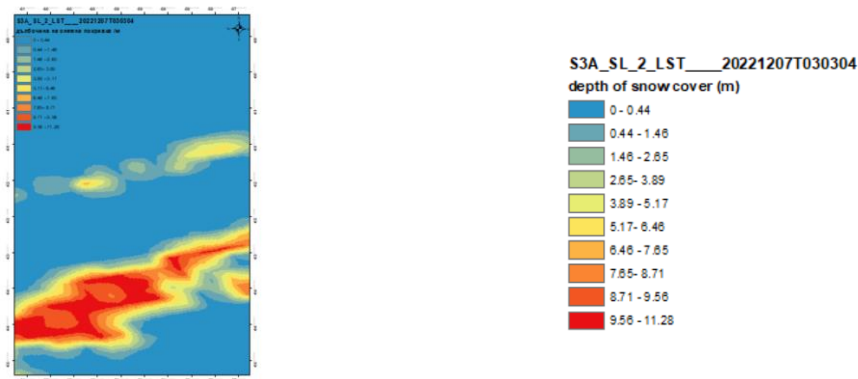


Figure 6 Depth of snow cover from Sentinel 3 SLSTR (07.12.2022)^{11, 15}

2.2 Data processing

The ecological monitoring of Livingstone Island, Antarctica has been done over a period of five years, from 2019 to 2023. Satellite images from the same period each year were used for this purpose. The selected satellite scenes were captured during the summer season, when the vegetation in the area is well represented, there is strong melting of wet snow, snow and ice, there are exposed rock massifs, strong presence of sea ice and ice soup.

For this purpose, four hybrid models were created for monitoring individual objects (figure 7- 10).

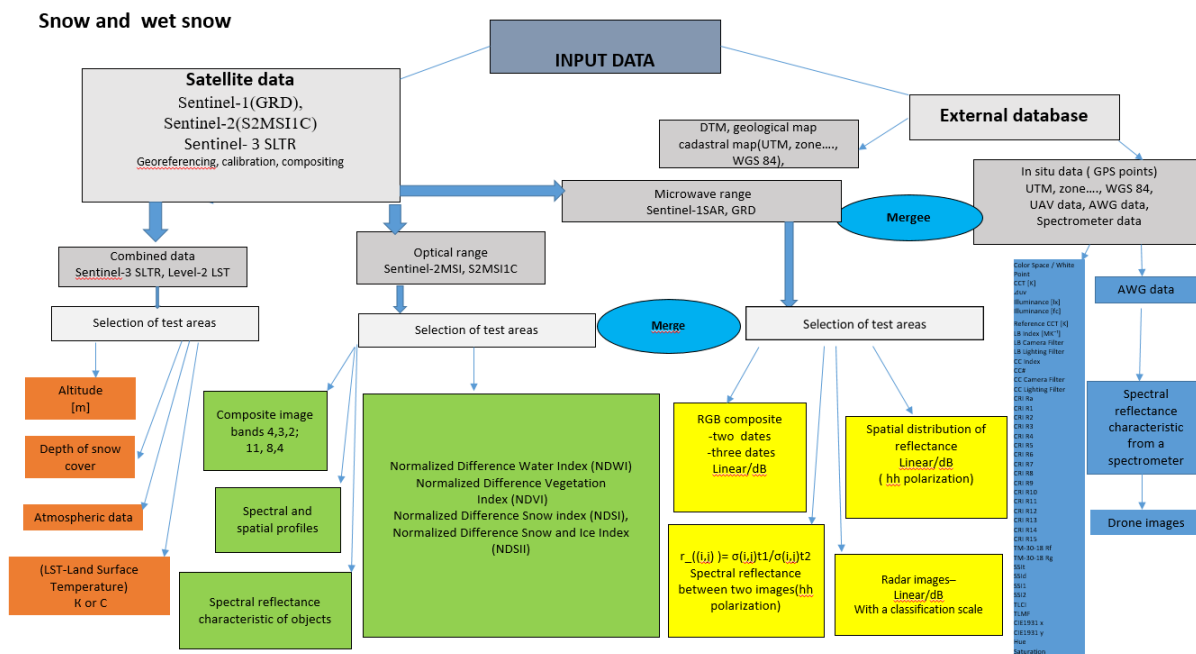


Figure 7 Hybrid model of methodologies for monitoring: Snow and wet snow

Ice, sea ice, permafrost and rocks

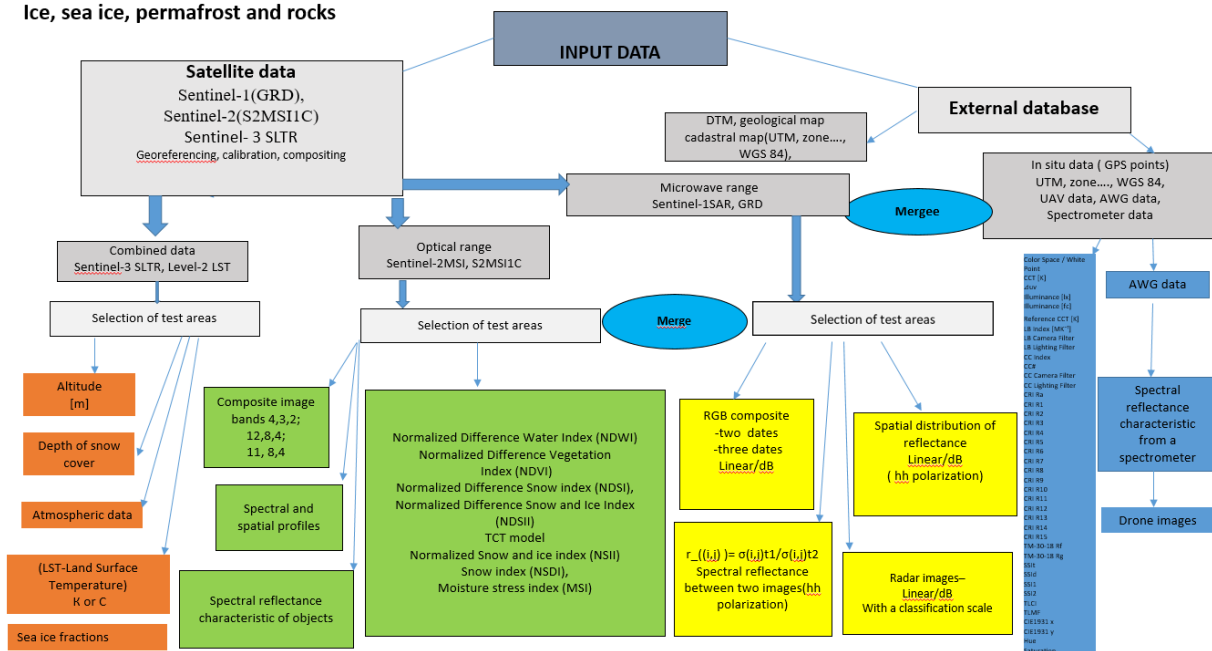
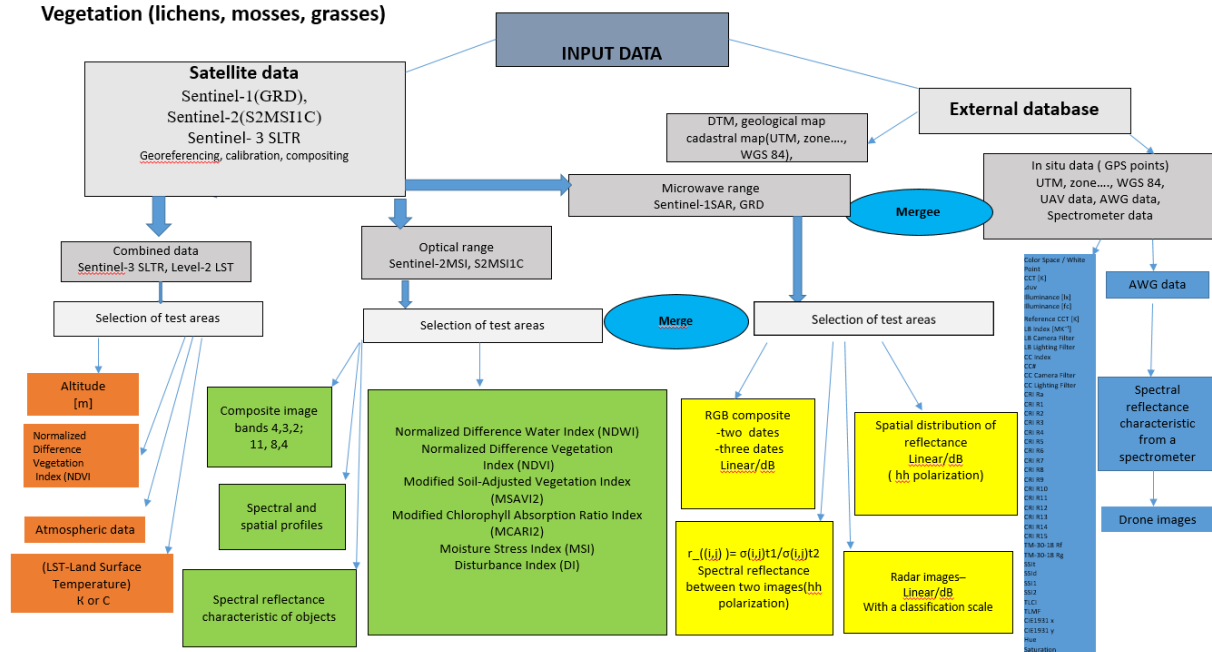


Figure 8 Hybrid model of methodologies for monitoring: Ice, sea ice, permafrost and rocks

Vegetation (lichens, mosses, grasses)



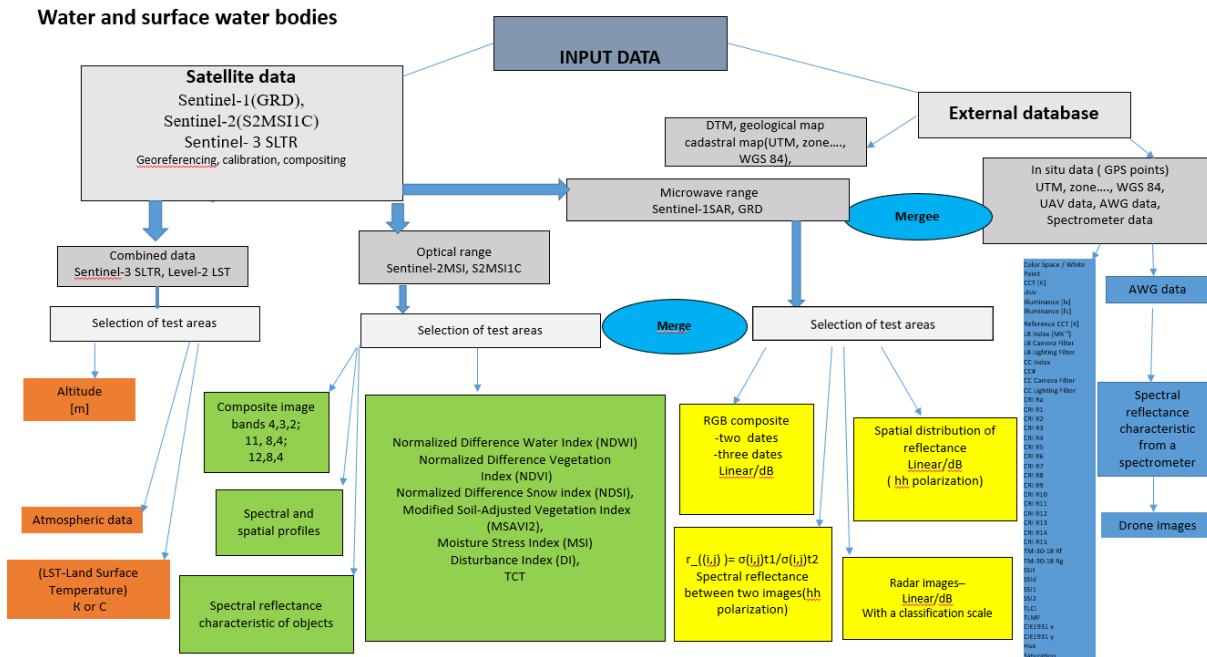


Figure 10 Hybrid model of methodologies for monitoring: Water and surface water bodies

3. RESULTS

As a result of the created hybrid models of monitoring methodologies, they can be used for weekly, monthly and annual monitoring. Some of the described metrics or indicators can also be used in the models according to the observed objects of interest and the availability of data.

Field research was done using a spectrometer on two territories of Livingston Island: Cape Hana Point and the territory around the Bulgarian Antarctic base (BAS). Over 100 spectral field profiles of various sites have been made, which can be found in machine-readable format on the official project page.

The purpose of this field study is to perform a spectral analysis and obtain representative statistical data of the most important objects of interest around the BAB.

A Sekonic C-800-U mobile spectrometer was used for field research.

The device measures natural light and flash, from 380 to 780 nm, a wide measurement range of color temperature (1600 to 40,000 K) and illuminance (1 to 200,000 lx in ambient light, 20 to 20,500 lx · s in flash).

Also measures: color rendering properties (CRI, SSI, TLCI, TLMF, TM-30-18), color control parameters (Hue/Saturation, CIE 1931 x, y chromaticity coordinate), lighting filter for Rosco© and LEE ©, camera filter for Kodak Wratten ©, LEE© and Fuji© and LB/CC index

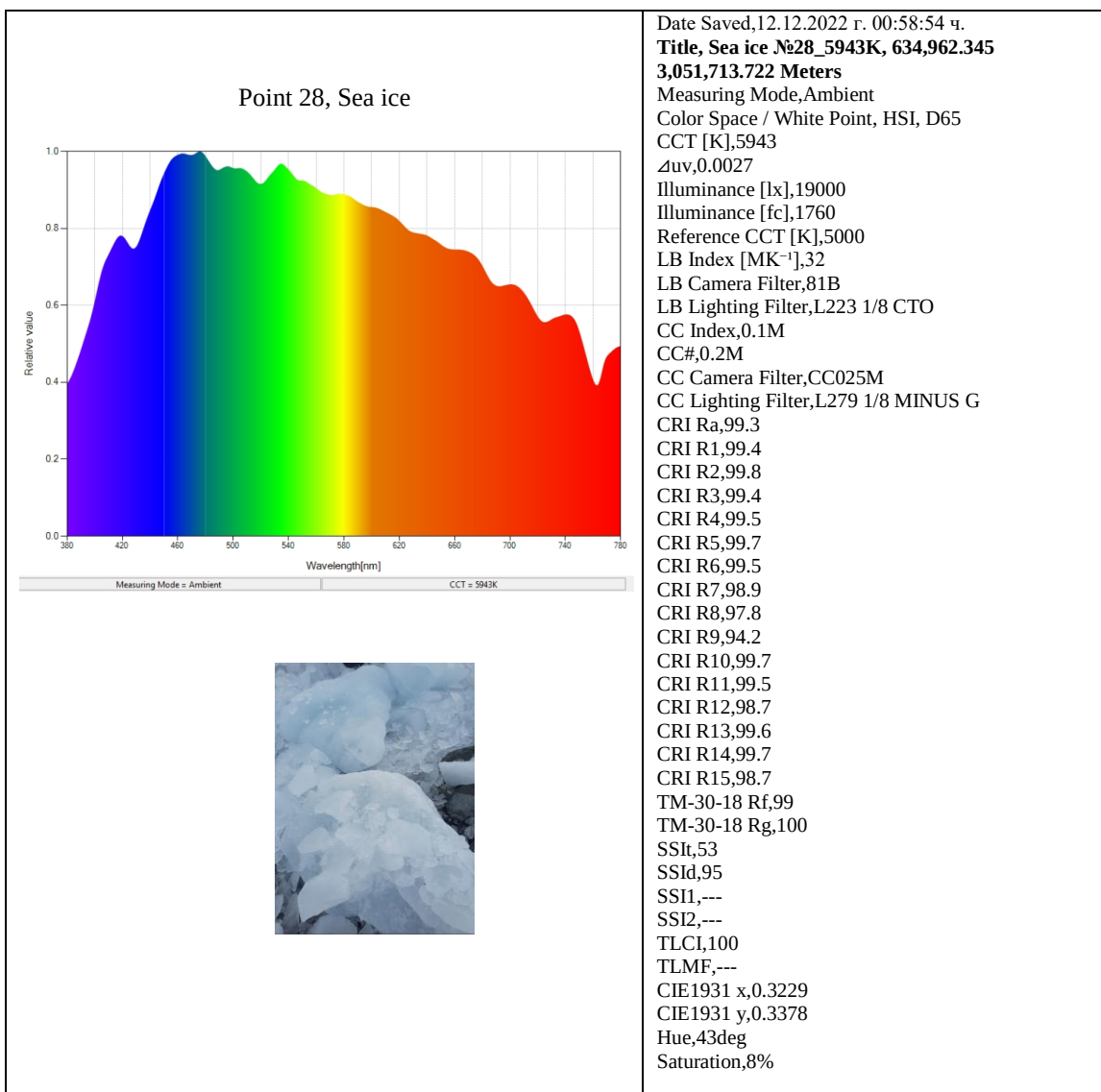
The C-800 Series Utility software can be freely downloaded from Sekonic's official website¹⁷.

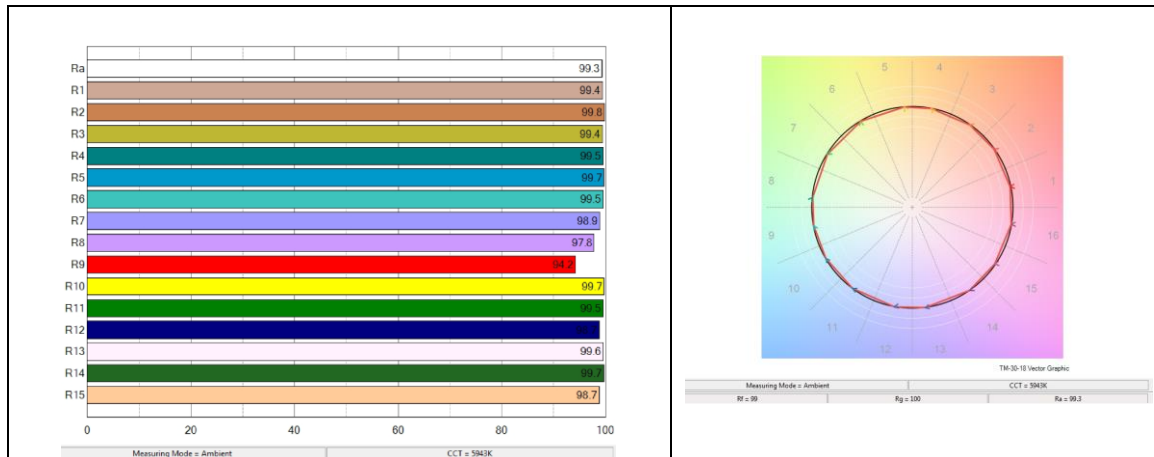


Figure 11 Sekonik C-800 spectrometer, archive на Temenuzhka Spasova

The table 4 demonstrates the data that was generated by the spectrometer. For each of the profile points, a photo was taken and GPS coordinates were taken.

Table 4 Data profile N28 from Sekonik C-800 Spectrometer.





The high color temperature values of the object that is demonstrated in fig. are 5943 in Kelvin. They ensure that the research is done in daylight and in clear weather. Also the CRI values are high. A CRI equal to 100 Ra is assumed to be daylight (sunlight), and in this particular case it is equal to 99.3 or extremely high. At this value of light, the colors of the observed objects are perceived most naturally. The same applies to the other values of R1 to R15, which for the object in the case of sea ice are on average 99. These values are also observed for the other objects.

The Ra value is the fraction of light produced by the light source. When measuring the luminous efficiency of a light fixture, in this case the Sun, CRI measures all color ranges from R1 to R15, but the RA value measures only the blue-green areas of the light fixture or sunlight, R1 to R8, at that Ra point = 99.3. The most difficult color to reproduce is red (R=9) and in the particular example it has the lowest values R9=94.2.

Data from the polar tests of the AWG station powered by the OXYMET™ environmental battery. Opportunities to use environmentally friendly renewable energy sources in the Antarctic even if only to power the seasonal needs of scientists and logisticians in the polar latitudes is an innovation, as enough energy is generated. The data are from 12/12/2022 (table 5). AWG coordinates are used for reference point.

Only UV index data was not generated, but this can be compensated with Sentinel 3 data.

The values for field survey days are positive, which in the last few years for this part of the year is similar to data from other BAB measuring instruments^{18, 19}.

Table 5 Data from Automatic recording weather station (AWG)³

Antarctic Weather and GPS logger - AWG "Alissa", powered by "OXYMET" - an ecological Magnesium-Air battery, design & development: Iliyan Popov, Boris Shirov							
GPS Lat.	GPS Lon.	Cell 1, Volts	Cell 2, Volts	Cell 3, Volts	Luminance	UV Index	Temperature, Celsius degree
-62.64	-60.37	1.12	2.1	3.06	4.97	0	9.2
-62.64	-60.37	0.9	1.93	3.26	4.96	0	9.22
-62.64	-60.37	1.04	1.8	3.15	4.97	0	9.4
-62.64	-60.37	1.04	2.12	3.13	4.96	0	9.6
-62.64	-60.37	0.95	1.96	3.35	4.97	0	9.65
-62.64	-60.37	1.18	2.23	3.15	4.97	0	9.88
-62.64	-60.37	1.17	2.12	3.4	4.97	0	9.81

-62.64	-60.37	0.98	2.16	3.34	4.97	0	9.9
-62.64	-60.37	0.98	1.96	3.14	4.97	0	9.97
-62.64	-60.37	1.06	1.9	3.29	4.96	0	10.02
-62.64	-60.37	1.13	2.1	3.63	4.96	0	10.06
-62.64	-60.37	1.14	2.15	3.25	4.97	0	9.96
-62.64	-60.37	1.07	2.4	3.59	4.97	0	10.08
-62.64	-60.37	1	2.12	3.15	4.96	0	10.1
-62.64	-60.37	1.04	2.15	3.22	4.97	0	10.14
-62.64	-60.37	1.16	2.34	2.95	4.96	0	10.17
-62.64	-60.37	1.07	2.2	3.2	4.97	0	10.21
-62.64	-60.37	1.14	2.39	3.33	4.97	0	10.22
-62.64	-60.37	1.09	2.16	3.27	4.96	0	10.24

Data from a drone was used to verify the current state of the investigated objects. The shots are from the dates 12.12, 18.12 and 24.12.2022, when a synchronous approach was made (figure 12).

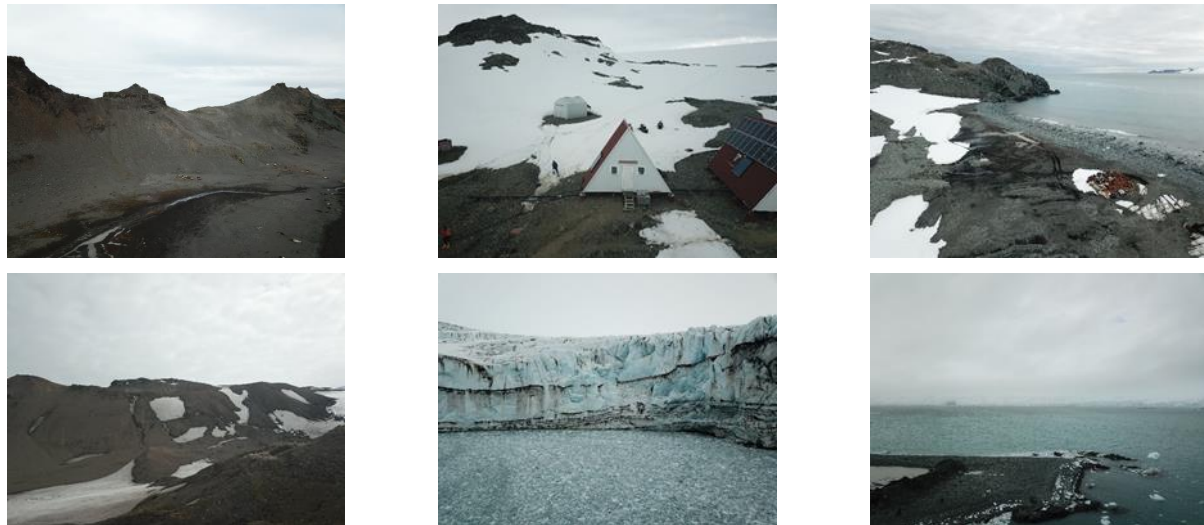
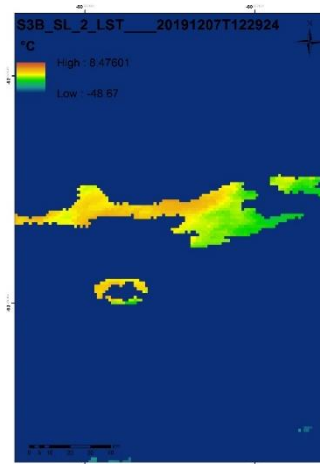
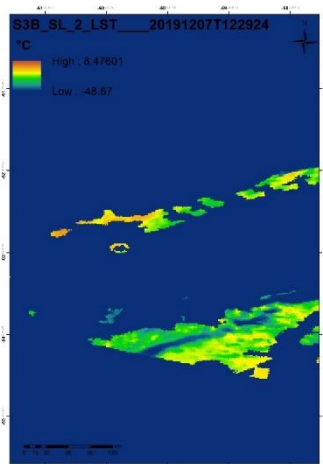
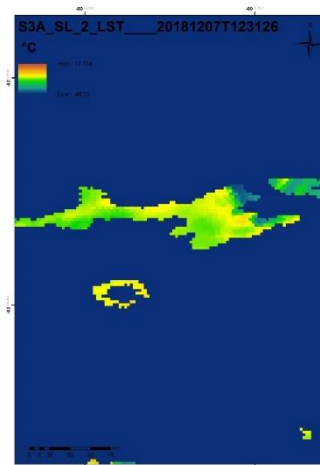
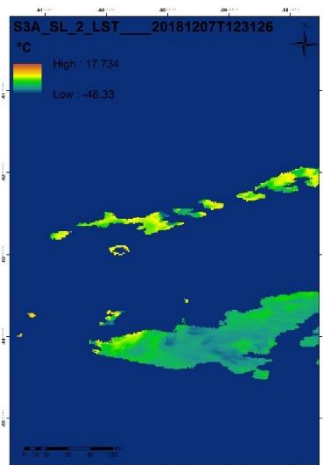


Figure 12 Dron images Author: Temenuzhka Spasova, December 2022, 31st Bulgarian Antarctic Expedition

Drone data was used to verify the current state of certain objects during field research and to create spectral profiles from satellite data (fig.12).

Sentinel 3 LST data were used despite the lower resolution but high temporal resolution over a period of five years to allow a comparison of five identical seasons (figure 13). A clear change of LST is observed in the Antarctic region and the BAB region in December 2018 and 2019, but it is not only on the database of 07.12, images from two more dates 12 and 24 December are considered, which change is significant during of the Antarctic summer of 2020. And since this is a snapshot reading, at that very moment a lower temperature was measured in the BAB region, but two weeks later the trend is exactly the same as for 2018 and 2019 and the values are clearly visible in figure 14.

There is an upward trend in the lowest temperature for the day, and also the highest, as in some places it reaches over 47 degrees by 2020, and after that the highest temperature decreases, but the lowest values increase sharply for the day and reach over 27 degrees.



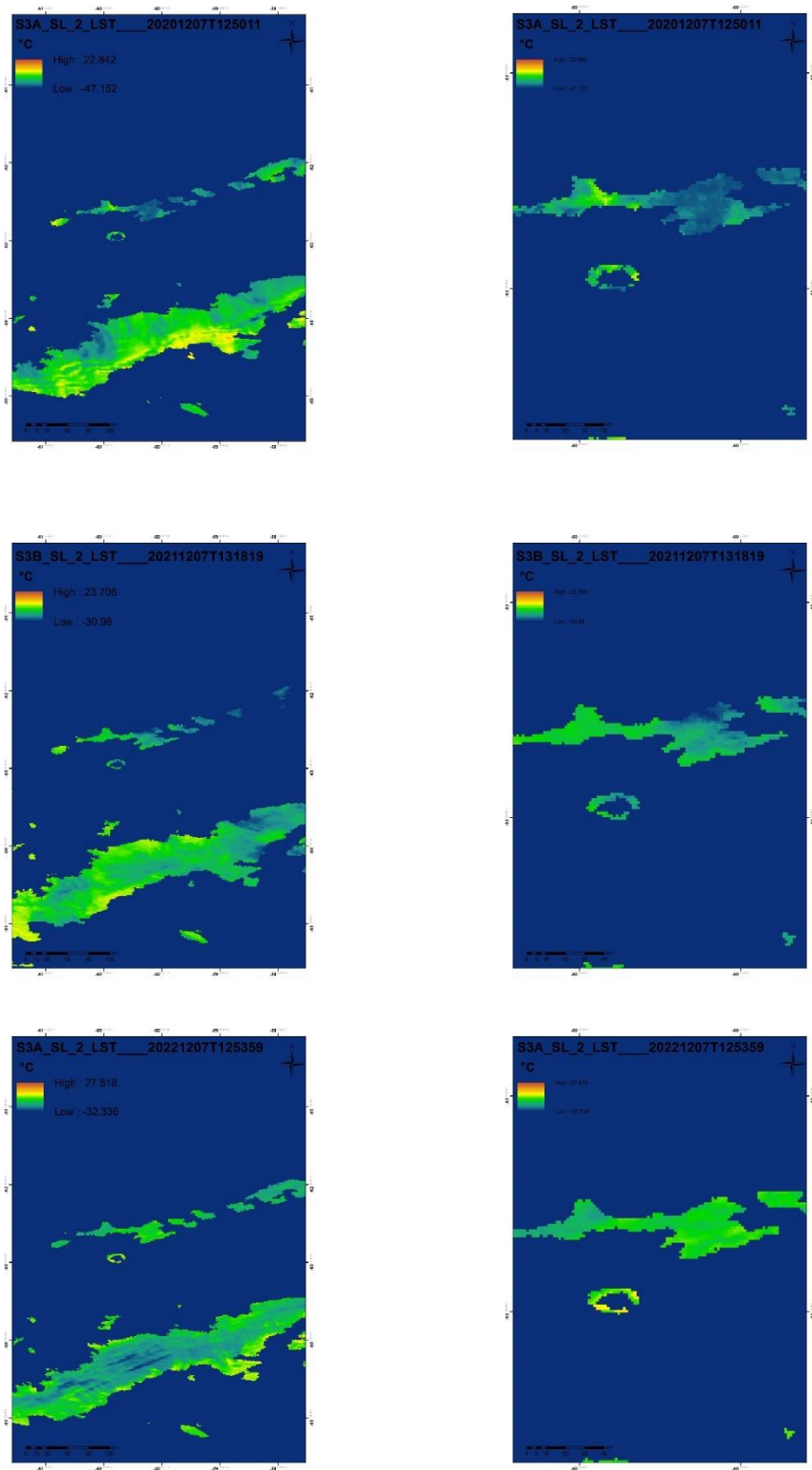


Figure 13 LST from Sentinel 3^{11, 15} Antarctica and Livingston Island on 07.12 for 5 years (2018 -2022)

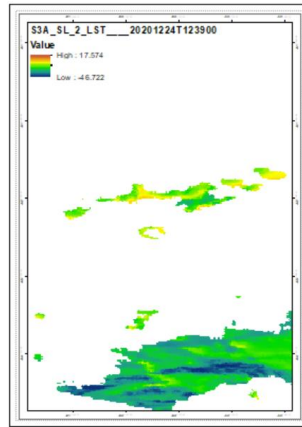


Figure 14 LST from Sentinel 3^{11, 15} Antarctica and Livingston Island on December 24, 2020

On the basis of LST, the lowest values of depth of snow cover from Sentinel 3 SLSTR (07.12.2022) were also measured, which is also visible from the footage on the ground, since on 07.12.2022 on Livingston Island around BAB there is a noticeable lack of wet snow and snow cover⁸. However, these values are directly inversely proportional to the areas with the presence of mosses, lichens and grasses.

From the made over 100 profiles, 8 points are demonstrated (fig. 15, 16, table 6), which are compared with the ground data from a spectrometer and for which it can be said that in the visible range from 380 to 780 μm the spectral curves overlap (figure 15, 16 and table 4).

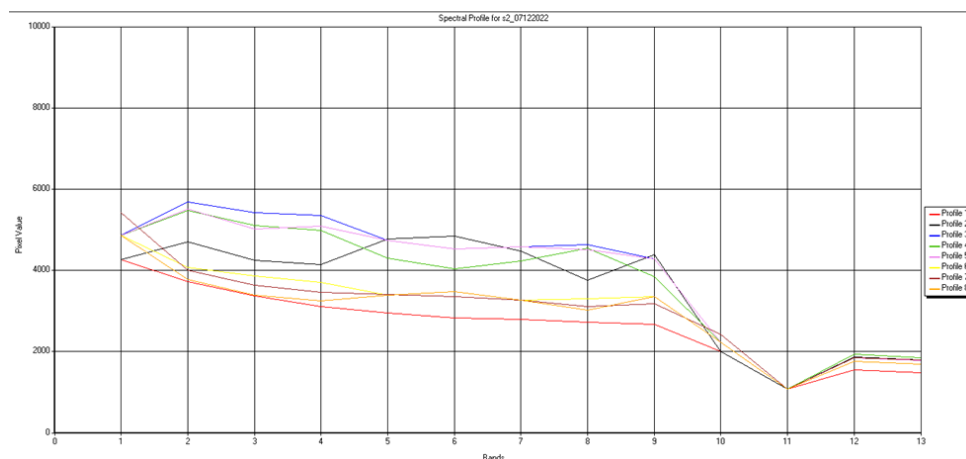


Figure 15 P1- rock on the shore, item 30; P2-sea ice from the coast, item 28 (fig. 1), P3- ice, item 29; P4-dry fine sand, item 23; P5- water (shallow river on the shore), item 25; P6-snow (dark), item 26; P7- fine sand, item 22; P8- water (river)



Figure 16 Spectral profile of points from Sentinel 2 MSI, 07/12/2022

Table 6 Spectral profile of points from Sentinel 2 MSI, 07/12/2022, CSV format

Layer	X Axis Values	Profile 1	Mean 1	Min 1	Max 1	Stddev Above 1	Stddev Below 1	Profile 2	Mean 2	Min 2	Max 2	Stddev Above 2	Stddev Below 2
1	1.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00	4263.00
2	2.00	3718.00	3696.56	3583.00	3806.00	3863.01	3572.99	4695.00	4729.00	3919.00	6320.00	6433.54	2996.46
3	3.00	3380.00	3361.22	3267.00	3452.00	3499.59	3260.41	4255.00	4414.00	3624.00	5977.00	5874.85	2635.15
4	4.00	3117.00	3107.33	2965.00	3251.00	3302.46	2931.54	4140.00	4292.89	3401.00	6047.00	5958.27	2321.73
5	5.00	2959.00	3041.67	2959.00	3211.00	3143.79	2774.21	4774.00	4071.89	3211.00	4774.00	6127.39	3420.61
6	6.00	2825.00	2934.22	2825.00	3120.00	3046.90	2603.10	4847.00	4071.78	3120.00	4847.00	6353.30	3340.70
7	7.00	2789.00	2906.67	2789.00	3154.00	3047.64	2530.36	4480.00	3906.67	3154.00	4480.00	5593.11	3366.89
8	8.00	2718.00	2711.22	2536.00	2901.00	2947.87	2488.13	3758.00	3850.89	3095.00	5290.00	5275.40	2240.60
9	9.00	2670.00	2786.00	2670.00	3024.00	2918.96	2421.04	4385.00	3800.22	3024.00	4385.00	5524.23	3245.77
10	10.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00	2008.00
11	11.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00	1081.00
12	12.00	1543.00	1647.44	1543.00	1879.00	1775.42	1310.58	1863.00	1901.67	1863.00	1988.00	1966.47	1759.53
13	13.00	1475.00	1571.67	1475.00	1783.00	1687.67	1262.33	1793.00	1821.00	1783.00	1892.00	1878.71	1707.29

Layer	Profile 3	Mean 3	Min 3	Max 3	Stddev Above 3	Stddev Below 3	Profile 4	Mean 4	Min 4	Max 4	Stddev Above 4	Stddev Below 4	Profile 5	Mean 5	Min 5	Max 5	Stddev Above 5	Stddev Below 5
1	4852.00	4655.67	4263.00	4852.00	5441.00	4263.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00
2	5681.00	5217.44	4295.00	6320.00	6978.67	4383.33	5475.00	4716.33	3995.00	5681.00	6848.79	4101.21	5501.00	4796.67	3891.00	5681.00	6877.41	4124.59
3	5426.00	4911.56	3927.00	5977.00	6725.89	4126.11	5100.00	4428.00	3621.00	5426.00	6400.48	3799.52	5018.00	4523.00	3611.00	5426.00	6359.56	3676.44
4	5353.00	4881.22	3829.00	6047.00	6721.24	3984.76	4990.00	4350.89	3605.00	5353.00	6326.18	3653.82	5085.00	4490.44	3427.00	5353.00	6496.66	3673.34
5	4740.00	4520.44	3562.00	4774.00	5555.82	3924.18	4307.00	4091.33	3362.00	4740.00	5443.51	3170.49	4740.00	4190.67	3362.00	4740.00	6005.50	3474.50
6	4521.00	4390.44	3670.00	4847.00	5314.61	3727.39	4033.00	3944.78	3428.00	4521.00	4885.02	3180.98	4521.00	4068.22	3428.00	4521.00	5505.90	3536.10
7	4587.00	4361.67	3493.00	4587.00	5301.83	3872.17	4227.00	3985.00	3256.00	4587.00	5352.11	3101.89	4587.00	4066.67	3256.00	4587.00	5823.48	3350.52
8	4625.00	4329.56	3580.00	5290.00	5697.27	3552.73	4537.00	3851.78	3204.00	4625.00	5697.20	3376.80	4514.00	3952.78	3131.00	4625.00	5686.59	3341.41
9	4275.00	4114.67	3470.00	4385.00	4910.98	3639.02	3846.00	3748.22	3226.00	4275.00	4651.87	3040.13	4275.00	3857.11	3226.00	4275.00	5173.74	3376.26
10	2231.00	2156.67	2008.00	2231.00	2454.00	2008.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00
11	1080.00	1080.33	1080.00	1081.00	1081.00	1075.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00
12	1856.00	1888.00	1856.00	1988.00	1951.89	1760.11	1927.00	1843.11	1705.00	1927.00	2119.19	1734.81	1856.00	1833.22	1705.00	1927.00	2009.98	1702.02
13	1782.00	1812.00	1782.00	1892.00	1864.96	1699.04	1851.00	1774.89	1655.00	1851.00	2021.65	1680.35	1782.00	1764.11	1655.00	1851.00	1919.05	1644.95

Layer	Profile 6	Mean 6	Min 6	Max 6	Stddev Above 6	Stddev Below 6	Profile 7	Mean 7	Min 7	Max 7	Stddev Above 7	Stddev Below 7	Profile 8	Mean 8	Min 8	Max 8	Stddev Above 8	Stddev Below 8
1	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	5426.00	5234.67	4852.00	5426.00	6000.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00	4852.00
2	4075.00	4313.67	3768.00	5001.00	5462.86	2687.14	4000.00	4051.33	3814.00	4425.00	4383.20	3616.80	3768.00	3965.11	3768.00	4215.00	4122.03	3413.97
3	3861.00	3975.11	3384.00	5100.00	5169.22	2552.78	3633.00	3720.67	3485.00	4104.00	3985.12	3280.88	3397.00	3648.22	3384.00	3962.00	3812.82	2981.18
4	3703.00	3866.44	3152.00	5085.00	5143.97	2262.03	3457.00	3585.00	3334.00	4018.00	3874.11	3039.89	3247.00	3499.00	3152.00	3929.00	3728.32	2765.68
5	3390.00	3785.67	3362.00	4740.00	4630.66	2149.34	3408.00	3728.56	3390.00	4430.00	4391.35	2424.65	3390.00	3643.89	3362.00	4295.00	4204.58	2575.42
6	3473.00	3758.11	3426.00	4521.00	4417.10	2538.90	3362.00	3701.33	3362.00	4425.00	4296.53	2427.47	3473.00	3624.56	3426.00	4088.00	4009.22	2596.78
7	3271.00	3666.33	3256.00	4697.00	4489.91	2052.09	3267.00	3605.56	3267.00	4352.00	4296.33	2247.67	3271.00	3514.67	3256.00	4136.00	4047.40	2494.60
8	3302.00	3501.56	2928.00	4537.00	4540.78	2063.22	3114.00	3228.33	3030.00	3572.00	3468.45	2759.55	3015.00	3180.56	2928.00	3477.00	3370.08	2659.92
9	3348.00	3582.22	3226.00	4275.00	4214.80	2481.20	3182.00	3511.89	3182.00	4121.00	4023.19	2340.81	3348.00	3467.56	3226.00	3941.00	3905.84	2790.16
10	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00	2231.00
11	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1081.33	1080.00	1082.00	1084.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00	1080.00
12	1758.00	1786.78	1705.00	1927.00	1909.55	1606.45	1852.00	1786.56	1663.00	1852.00	1988.60	1715.40	1758.00	1706.56	1591.00	1758.00	1875.77	1640.23
13	1696.00	1723.22	1655.00	1851.00	1829.25	1562.75	1789.00	1726.67	1600.00	1789.00	1921.94	1656.06	1696.00	1649.89	1555.00	1696.00	1802.16	1589.84

Through the SAR data, a warming trend is observed in the regions of interest, which demonstrate low dB reflectance values, which is a sure sign of heavy melting and the presence of water even over areas normally occupied by snow and ice (figure 17). January 2021, January and February 2022 and January 2023. The year 2022 makes an extremely big impression. As the surface is highly melted throughout the month of February, which is not surprising because a similar trend is observed for December 2022, when we have serious increases in LST at the lowest temperatures.

SAR gives information about the presence of wet snow, ice and helps to identify ice rocks, as there is already a proven classification of objects ^{8, 13}.

The results, which were used to carry out a complex approach by the author, are from a 5-year period and coincide with a period in which field research was done, and for data verification and validation, multiple spectral profiles were made using a spectrometer. In this way, the spectral curves can be compared with those of a spectrometer, the data from a meteorological battery can be used, and also the data from every single instrument located on Livingstone Island, Antarctica, in order to obtain very comprehensive information, rather than interpolation data that averages the values of different coefficients.

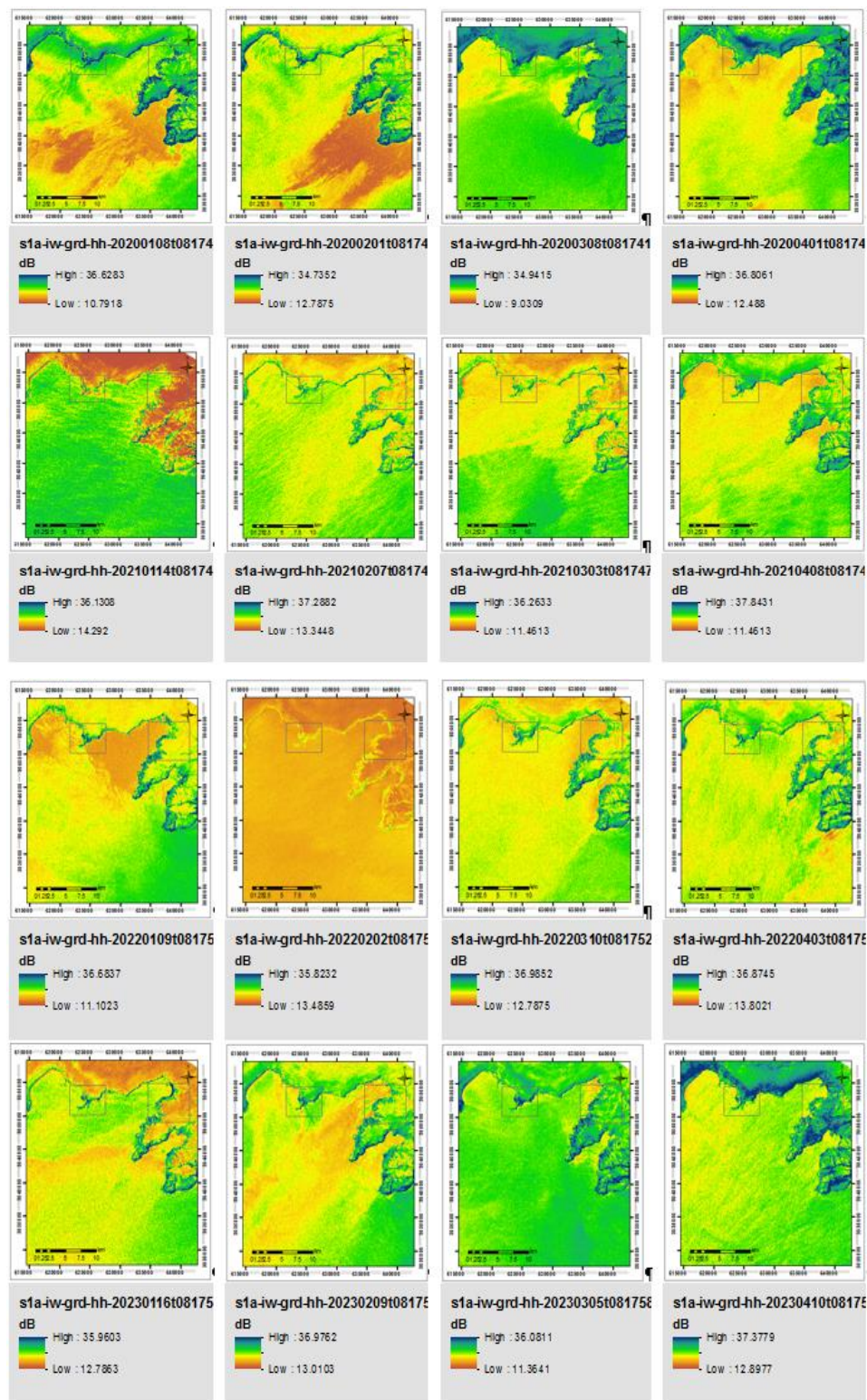


Figure 17 SAR images from Sentinel 1 in dB

4. CONCLUSION

Spectral curves have long shown their benefits, as they do not change, but can only serve to validate research in polar regions, as very often research is seasonal, and in winter the sites are difficult to access.

Seasonal field surveys are used to supplement and enrich information on spectral profiles and libraries for specific sites or a group of sites.

Regardless of the observed objects, it's desirable first to match data from optical and radar images, because if there is specific data that is not visible with one satellite, it can be compensated by a second or third. This necessitates the creation of a data space in which data can be processed, but also processed by different experts, in order to extract maximum information about the studied area. And last but not least, that this information is similarly structured so that it can be shared and interoperable.

It turned out that the dynamics of Livingstone Island can be successfully inferred by the three satellites Sentinel 1 SAR, Sentinel 2 MSI and Sentinel 3 LSTR, and also individually, but only with the availability of quality optical range data.

The parameters obtained from Sentinel 3 LSTR are absolutely sufficient for the initial needs of building polar space data for initiatives such as Digital Twins, Oceans and Seas with an average terrain resolution of 1 km, but periodically it will be good to verify and with data from the other two satellites, and why not using more satellites like Sentinel 5P.

The four hybrid remote sensing monitoring models thus constructed could be successfully applied to the various needs, as they have already been proven to work through a synchronous approach during the 31st Bulgarian Antarctic Expedition. With the models thus distinguished, we could partially support the structuring of the data on specific topics or objects of interest.

And since there is also data and research from previous years, which are based on open data from the Copernicus program⁸, the following conclusions can be drawn:

1. The dynamics of temperature change from 2018 to 2022 are clearly visible, and for this purpose, many more images than the ones presented were used.
2. In addition to the temperature, the changes in the presence of snow cover from Sentinel 2, the retreat of the ice cover, and the lack of sea ice characteristic of the area are also clearly observed.
3. All spectral indices can be successfully applied when there is a lack of snow cover and rock exposure. There, the spectral profiles show a 100% match between in situ and satellite data. This helps the good study of cracks in the rocks, which are additionally verified with the radar data, and where there are ground points, it is even recommended to use them together with drone footage and photogrammetric models.
4. SAR continues to be the most accurate indicator when selecting data, but Sentinel 3 gives us daily information on climate change, which also gives enough additional parameters
5. From the conducted research and what has been seen on the ground, it can be concluded that the territory of BAB, Cape Hanna Point and large areas of the western part of the island in the last few years have been formed as local small heat islands for Antarctica.

As for Hanna Point due to orographic reasons and the accumulation of large colonies of polar species of animals, this phenomenon has been observed in previous years and is also visible in the large amount of vegetation such as mosses, lichens and grasses.

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